MIL-C-3737
17 APRIL 1952

MILITARY SPECIFICATION

CALIPER, OUTSIDE (OPHTHALMIC LENS)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers calipers (degree gauge) suitable for use in measuring the thickness of ophthalmic lens.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-601 — Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 — Boxes, Wood, Nailed and Lock-Corner.
NN-B-631 — Boxes, Wood, Wirebound (for Domestic Shipment).

MILITARY

JAN-P-105 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
MIL-B-107 — Boxes, Wood, Wirebound (Overseas Type).
JAN-P-120 —Packaging and Packing for Overseas Shipment — Cartons, Folding, Paperboard.
JAN-P-125 —Packaging and Packing for Overseas Shipment—Barrier Materials, Waterproof, Flexible.
JAN-P-133 —Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.
MIL-A-140 — Adhesive, Water-Resistant, Waterproof Barrier-Material.

STANDARDS

MILITARY

MIL-STD-105— Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking of Shipments.

DRAWINGS

Armed Services Medical Procurement Agency

1742—Calipers, Thickness.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Steel.*—The calipers, except the quadrant and spring, shall be made of S.A.E.1010 steel, having a Rockwell hardness of 90 on the "B" scale.

3.1.2 *Brass.*—The quadrant shall be made of brass.

3.1.3 *Spring steel.*—The spring portion shall be made of suitable spring steel.

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3.2 **Style and design.**—Style and design shall conform to figure 1, and shall be thickness calipers (degree gauge), with vernier attachment, and with adjustable point and adjustable set screw; shall measure ophthalmic lens thickness in tenths of millimeters, from 0 to at least 13 millimeters, and the scale shall be graduated to at least 12 mm.

3.2.1 *Construction.*—The calipers shall be so constructed as to insure durability, accuracy, and ease in handling and reading.

3.3 **Finish.**—The calipers shall be nickel plated and be given a satin finish.

3.4 **Identification markings.**—Each instrument shall be legibly and permanently marked with the name of the manufacturer or registered trade-mark.

3.5 **Workmanship.**—Workmanship shall be first class throughout. Calipers shall be free from defects which detract from their appearance or impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.1). In addition, inspection at contractor's facility by inspector shall be conducted in accordance with Standard MIL-STD-105.

4.2 **Inspection.**—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery, as specified (see 6.1). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 **Tests.**—Tests shall be conducted in accordance with the applicable specifications, as necessary, to determine compliance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit packaging.*—One caliper constitutes one unit. Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification JAN-P-120 or JAN-P-133. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.2 *Intermediate packaging.* — Twelve unit cartons (12 units) either shall be packaged in a carton of appropriate size and design, or be completely wrapped with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.2 Packing.

5.2.1 *For domestic shipment.*—Sixteen intermediate cartons (192 units) shall be overpacked in an exterior container constructed in accordance with Specification NN-B-601, NN-B-621, or NN-B-631.

5.2.2 *For oversea shipment.*—Sixteen intermediate cartons (192 units) shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with Specification JAN-P-105, JAN-P-106, or MIL-B-107. Closure and strapping shall comply with appendix of applicable specification. Each box shall be lined with a waterproof barrier (bag) conforming to Specification JAN-P-125. Closure shall be completely sealed by a continuous seam of water-resistant adhesive at least three-quarter inch wide and conforming to Specification MIL-A-140.

5.2.3 *Packing variation permitted.*—If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in an exterior container of suitable size and design and so constructed as to insure safe delivery to specified destination.

5.3 Marking.—Interior packages (unit or intermediate) as applicable and all shipping containers shall be marked in accordance with Standard MIL-STD-129 and ASMPA Marking Instructions.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following:

- (a) Title, number, and date of this specification.
- (b) Sampling required (see 4.1).
- (c) Inspection required (see 4.2).
- (d) Place of final inspection (see 4.2).
- (e) Whether packing for domestic or oversea shipment is required (see 5.2).

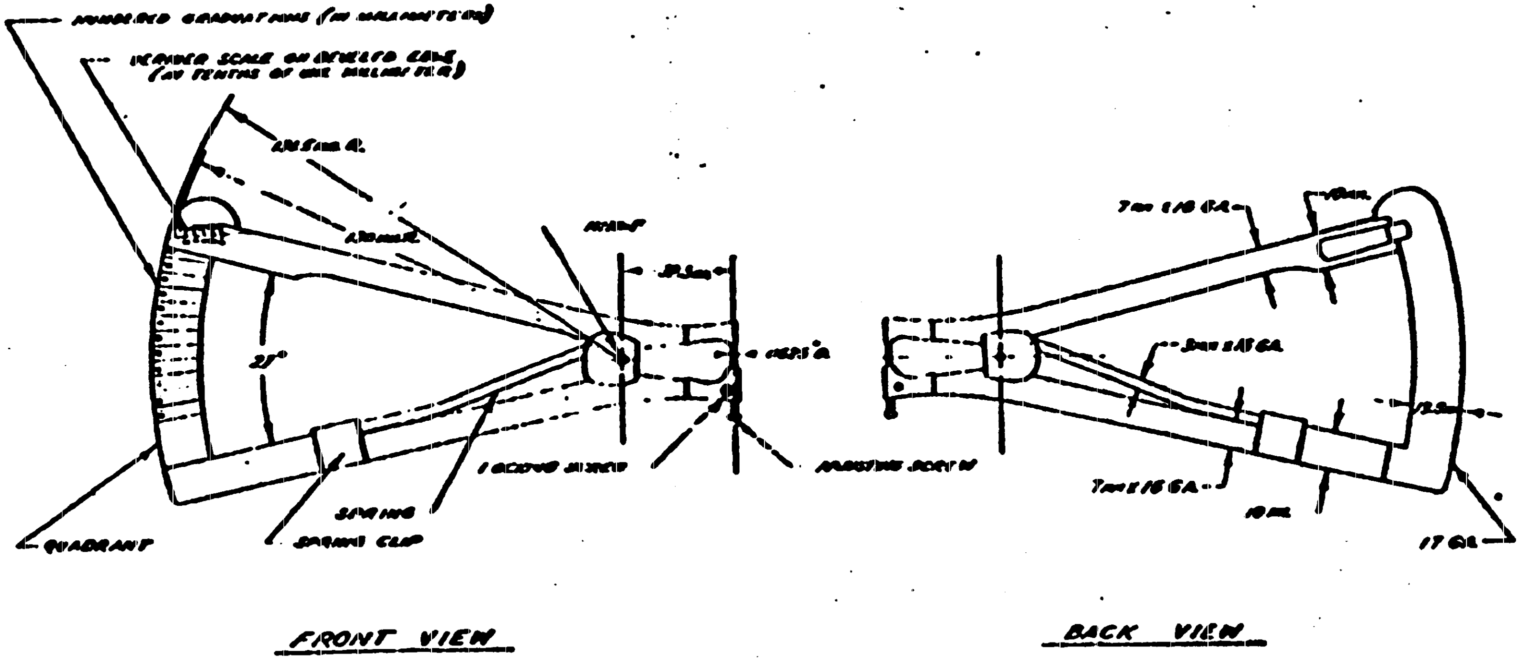
6.2 This specification covers the following item appearing in the Armed Services Catalog of Medical Matériel:

14-060-000—Calipers, Thickness.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Medical Service
Navy—Bureau of Medicine and Surgery
Air Force.


NUMBERED GRADUATIONS (IN MILLIMETERS)
VERNIER SCALE ON BEVELED EDGE (IN TENTHS OF ONE MILLIMETER)
QUADRANT
SPRING
SPRING CLIP
LOCKING SCREW
RIVET
MEASURING POINT
17 GA.
10 GA.
FRONT VIEW
BACK VIEW

NOTE: MAT'L QUADRANT BRASS
SPRING - SPRING STEEL
BALANCE OF CALIPERS - SAE 1010 STEEL, NICKEL PLATED, SATIN FINISH.

FIGURE 1.—Calipers, thickness.